Python(22AIE205) Programming Lab

Weekly Lab Report-1

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1. Questions Given

Question 1: Write a Python program to print the frequency of each character in a sentence.

Question 2: Write a Python program to find number of vowels and consonants in a sentence.

Question 3: Write a Python program to remove duplicates in a list.

Question 4: Write a Python program to find duplicates in a list.

Question 5: In a company, 2 teams attended different technical workshops. The sets team A and team B store the list of workshops attended by each team. Write a Python program to: 1) Workshops attended by both teams. 2) Workshops attended by only team A. 3) Workshops on unique workshops attended. 4) Workshops attended by only team B. 5) Workshops attended by only 1 team.

Question 6: Write a python program to find the Transpose of the given matrix.

Question 7: Write a python program to find if given two strings are anagrams or not.

Question 8: Create a cab booking system using tuples to store the cab details. Display the available cabs, accept the distance and car ID from the user, calculate the total fare, and confirm the booking.

Question 9: Accept the details of students along with their marks in three subjects. Calculate the total, average and grade of each student and store them in a nested dictionary and print the results.

2. Programs, Outputs and Results

Question-1

Aim: Write a Python program to print the frequency of each character in a sentence.

Code:

```
# Question 1
sentence = input("Enter a sentence: ")
a = {}
for ch in sentence:
    if ch in a:
        a[ch] += 1
    else:
        a[ch] = 1
for ch,count in a.items():
    print(f"'{ch}': {count}")
```

Output:

```
>>>
====== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\2lab5.py ======
Enter a sentence: pranav
'p': 1
'r': 1
'a': 2
'n': 1
'v': 1
>>>
```

Result: Read each individual character from the given sentence or a string and printed the frequency of each character.

Question-2

Aim: Write a Python program to find number of vowels and consonants in a sentence.

Code:

```
sentence = input("Enter a sentence: ")
v = ['a','e','i','o','u','A','E','I','O','U']
vowel = 0
consonant = 0
for i in sentence:
```

```
        if i in v:
            vowel += 1
        else:
            consonant += 1
    print("Vowels:",vowel,"\nConsonants:",consonant)
```

Output:

```
>>>
====== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\2lab5.py ======
Enter a sentence: apple
Vowels: 2
Consonants: 3
>>>
```

Result: Read the string from the user and counted the number of vowels and consonants.

Question-3

Aim: Write a Python program to remove duplicates in a list.

Code:

```
my_list = ["a","b","a",1,2,3,1,2,3,3]

unique = list(set(my_list))

print(unique)
```

Output:

```
>>>
====== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\2lab5.py ======
[1, 'a', 3, 2, 'b']
>>>
```

Result: Converted the given list into a set to remove the duplicate elements.

Question-4

Aim: Write a Python program to find duplicates in a list.

Code:

```
    new1 = [1,1,2,3,3,4,5]
    dup = []
    for i in new1:
        if new1.count(i)>1:
            dup.append(i)
    print(dup)
```

Output:

```
>>>
====== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\2lab5.py ======
[1, 1, 3, 3]
>>>
```

Result: Found the list of duplicate elements without using sets.

Question-5

Aim: In a company, 2 teams attended different technical workshops. The sets team A and team B store the list of workshops attended by each team. Write a Python program to find the specified shared, unique and team-specific workshops.

Code:

```
A = {'a','b','c'}
B = {'b','c','d'}
one = A & B
two = A - B
three = A | B
four = B - A
five = A or B
print(one,two,three,four,five)
```

Output:

```
>>>
====== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\2lab5.py ======
{'b', 'c'} {'a'} {'d', 'a', 'b', 'c'} {'d'} {'b', 'c', 'a'}
>>>
```

Result: Implemented Union, Intersection, Difference and Symmetric Difference.

Question-6

Aim: Write a python program to find the Transpose of the given matrix.

Code:

```
matrix = [[1, 2, 3],[4, 5, 6]]
transpose = []
for i in range(len(matrix[0])):
    new_row = []
    for row in matrix:
        new_row.append(row[i])
    transpose.append(new_row)
print(transpose)
```

Output:

```
>>>
====== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\2lab5.py ======
[[1, 4], [2, 5], [3, 6]]
>>>
```

Result: Using two for loops we exchanged the elements to find the transpose of the matrix.

Question-7

Aim: Write a python program to find if given two strings are anagrams or not.

Code:

```
a = input("Enter first string: ")
b = input("Enter second string: ")
if sorted(a) == sorted(b):
    print("Anagrams")
else:
    print("Not an Anagram")
```

Output:

```
>>>
====== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\2lab5.py ======
Enter first string: pranav
Enter second string: vishal
Not an Anagram
>>>
```

Result: Sorted the strings and compared them to find if they are anagrams or not.

Question-8

Aim: Create a cab booking system using tuples to store the cab details. Display the available cabs, accept the distance and car ID from the user, calculate the total fare, and confirm the booking.

Code:

```
cabs = [
    (101,"Raju","Bike",12),
    (102,"Ramu","Car",25),
    (103,"Ravi","Luxury",60)
]

# Display available services
dist = float(input("Enter trip distance: "))
cab_id = int(input("Enter cab ID: "))
total = 0
selected = False
for i in cabs:
    if i[0]==cab_id:
        total = dist*i[-1]
        selected = True
        break

if selected:
    print("\n-----Booking Summary-----\n")
    print("Cab ID:",cab_id,"\nDriver:",i[1],"\nVehicle:",i[2],"\nRate per
Kilometer:",i[-1],"\nTotal cost:",total)
else:
    print("Booking failed. Invalid Car ID")
```

Output:

```
>>>
====== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\2lab5.py ======
Enter trip distance: 40
Enter cab ID: 101

-----Booking Summary-----

Cab ID: 101
Driver: Raju
Vehicle: Bike
Rate per Kilometer: 12
Total cost: 480.0
>>>
```

Result: Read inputs from the user and displayed the appropriate message based on the cab id and calculated total fare/cost to travel.

Question-9

Aim: Accept the details of students along with their marks in three subjects. Calculate the total, average and grade of each student and store them in a nested dictionary and print the results.

Code:

```
report = {}
n = int(input("Enter number of students: "))
for i in range(n):
    name = input("Enter name of the student: ")
    sub = int(input("Enter number of subjects: "))
    marks_dict = {}
    for j in range(sub):
        sub_name = input("Enter subject name: ")
        sub_marks = int(input("Enter marks: "))
        marks_dict[sub_name] = sub_marks
    report[name] = marks_dict
print("\n---STUDENT REPORT CARD---\n")
for name,subjects in report.items():
    print("Student name:",name)
    total = sum(subjects.values())
    average = total/len(subjects)
    if average>=90:
        grade = "A+"
    elif average >=80:
        grade = "A"
    elif average >=70:
        grade = "B"
    elif average >=50:
        grade = "C"
    elif average >=35:
        grade = "D"
    else:
        grade = "F"
    print("Subjects and marks:")
    for subject,marks in subjects.items():
        print(f"{subject}:{marks}")
    print("Total marks:",total)
```

```
        print("Average marks:",average)
        print("Grade:",grade)
```

Output:

```
>>>
====== RESTART: C:\Users\prana\OneDrive\Desktop\AV.SC.U4AIE25102\2lab5.py =====
Enter number of students: 2
Enter name of the student: tej
Enter number of subjects: 2
Enter subject name: math
Enter marks: 90
Enter subject name: physics
Enter marks: 85
Enter name of the student: raj
Enter number of subjects: 3
Enter subject name: math
Enter marks: 87
Enter subject name: physics
Enter marks: 78
Enter subject name: biology
Enter marks: 89

---STUDENT REPORT CARD---

Student name: tej
Subjects and marks:
math:90
physics:85
Total marks: 175
Average marks: 87.5
Grade: A
Student name: raj
Subjects and marks:
math:87
physics:78
biology:89
Total marks: 254
Average marks: 84.66666666666667
Grade: A
>>>
```

Result: Successfully read details from the user and stored the details of the students and their respective marks using nested dictionaries.

3. Observations

• Sets and Dictionaries: Dictionaries make it simple to count individual character frequencies, while sets easily remove repeated items from lists.

• Set Operations: Using set operators like &, -, and | makes comparing shared and unique data between two groups fast and simple.

• String and Matrix handling: Built-in functions like sorted() make checking for anagrams easy, while nested loops allow quick matrix transposition.

• Tuples and nested dictionaries: Using tuples and nested dictionaries provides an easy way to organize structured real-world data like student report cards and cab booking details.

4. Overall Result

All assigned programs were implemented and tested successfully.

Student Signature: _____ Faculty Signature: _____